VIRTUAL PRODUCTION

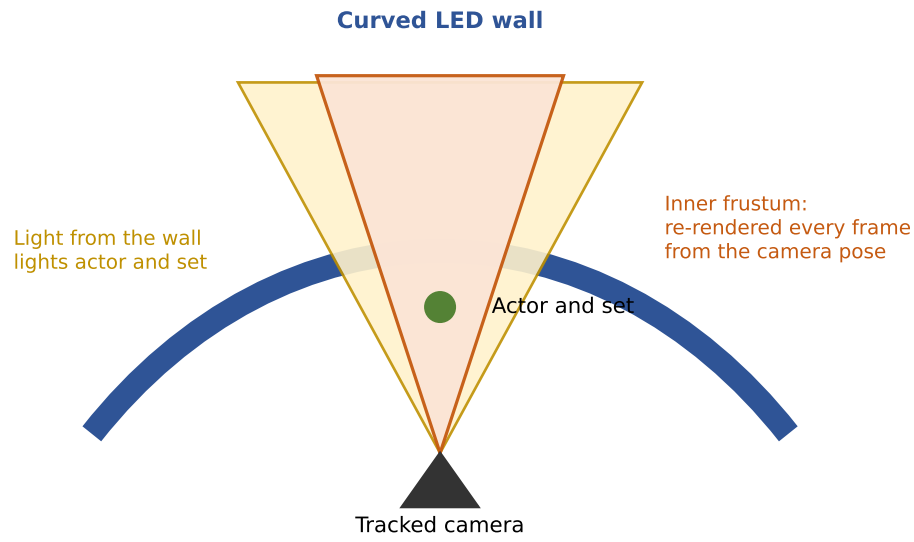
Real-time computer graphics on an LED stage

Computer Graphics | Semester V | CO1 to CO6

1. PROBLEM STATEMENT

- Green-screen filming hides the real set from actors and camera. Light and reflections on actors never match the final background, and the result is seen only weeks later in post-production.
- Shooting on real locations costs travel time and money, and weather and daylight cannot be controlled.
- Goal: show a realistic, correctly lit 3D background on set, in real time, drawn from the camera's own point of view.

HOW IT WORKS



2. GRAPHICS CONCEPTS USED

CO1 — **Coordinate systems and primitives**

World, camera and LED-wall coordinate systems. Scenes are built from triangles, lines and points.

CO2 — **Scan conversion and filling**

Rasterization turns triangles into pixels; the wall shows millions of pixels refreshed at film frame rates.

CO3 — **Transformations**

Tracked camera position and rotation give the view matrix; models move, turn and scale with 4x4 matrices.

CO4 — **Visible surfaces and shading**

Z-buffer hides surfaces; Blinn-Phong and image-based lighting give light, shadows and reflections.

CO5 — **Viewing, clipping, projection**

Off-axis perspective from the camera's position keeps parallax correct; clipping limits drawing to the visible region.

CO6 — **Curves, surfaces, display systems**

Curved LED volume, spline and subdivision surfaces for sets, GPU pipeline with frame synchronisation (genlock).

3. BENEFITS

- Actors see the environment and react naturally.
- Real light from the LED wall gives believable reflections and skin lighting.
- Directors see the shot on set and can change time of day or place in minutes.
- Less travel, less green-screen clean-up, fewer reshoots.
- One stage can be reused for many scenes.
- Limits: high cost, moire patterns, colour calibration, tracking delay.

4. FUTURE SCOPE

- AI and neural rendering (for example Gaussian splatting captures of real places) for photo-real backgrounds.
- Faster GPUs and cloud rendering for smaller studios and mobile stages.
- Digital twins of real locations, updated from scans.
- Real-time ray tracing and photo-real digital humans.
- Reuse of the same 3D assets in AR, VR and the metaverse.
- Lower-delay tracking and automatic colour calibration.

Virtual production = real-time rendering + camera tracking + correct projection.

References: T. Akenine-Moller et al., Real-Time Rendering, 4th ed., CRC Press, 2018. D. Hearn and M. P. Baker, Computer Graphics with OpenGL, 4th ed., Pearson, 2011.

Name: _____ Roll No: _____